

Algebra 1

Unit 5

Lesson 7 Homework

Name Answer Key

Date _____

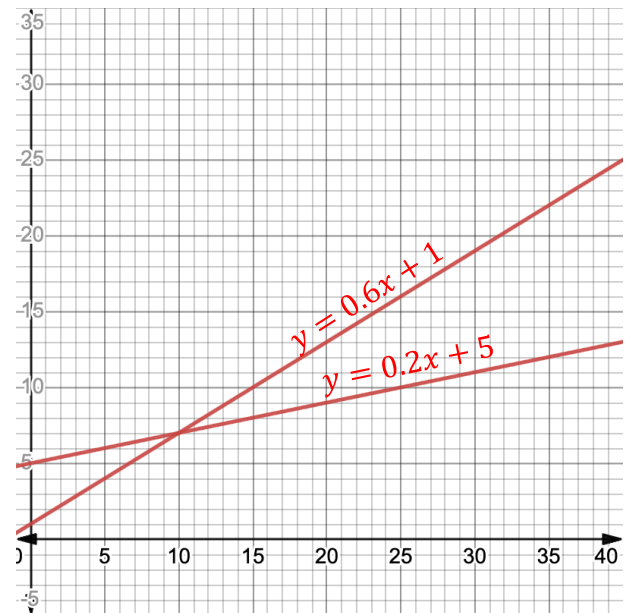
Period _____

Use the system of equations $\begin{cases} y = 0.2x + 5 \\ y = 0.6x + 1 \end{cases}$ to complete the following.

- Graph the system of equations on the coordinate plane.

- Complete the statement.

The point (10, 7) is the solution to the system.



Jake has a total of 12 quarters and dimes worth \$1.95. The system of equations $\begin{cases} d = 12 - q \\ d = 19.5 - 2.5q \end{cases}$ can be used to represent the amount of quarters, q , and the amount of dimes, d .

The graph of the system is shown.

- Label the axes.
- What are any possible constraints on the x -values?

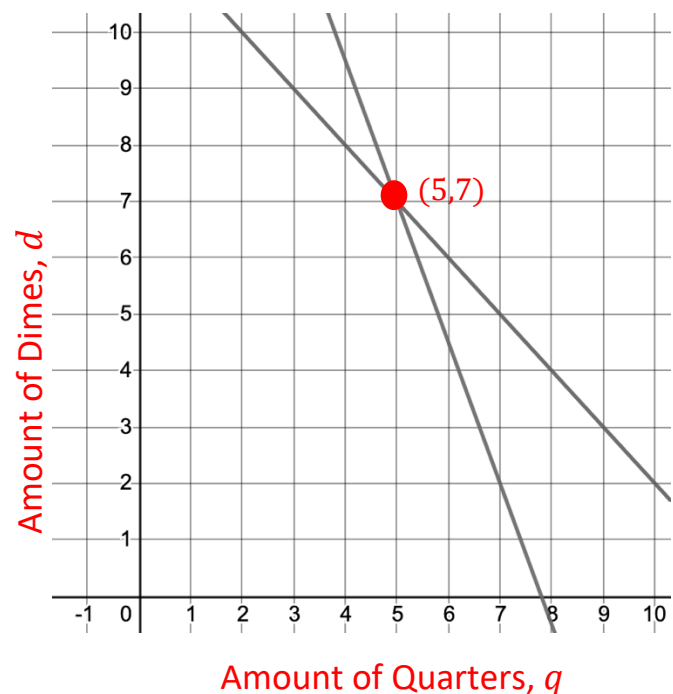
$$x \geq 0$$

- What are any possible constraints on the y -values?

$$y \geq 0$$

- What is the solution to this system of equations?

$$(5, 7)$$

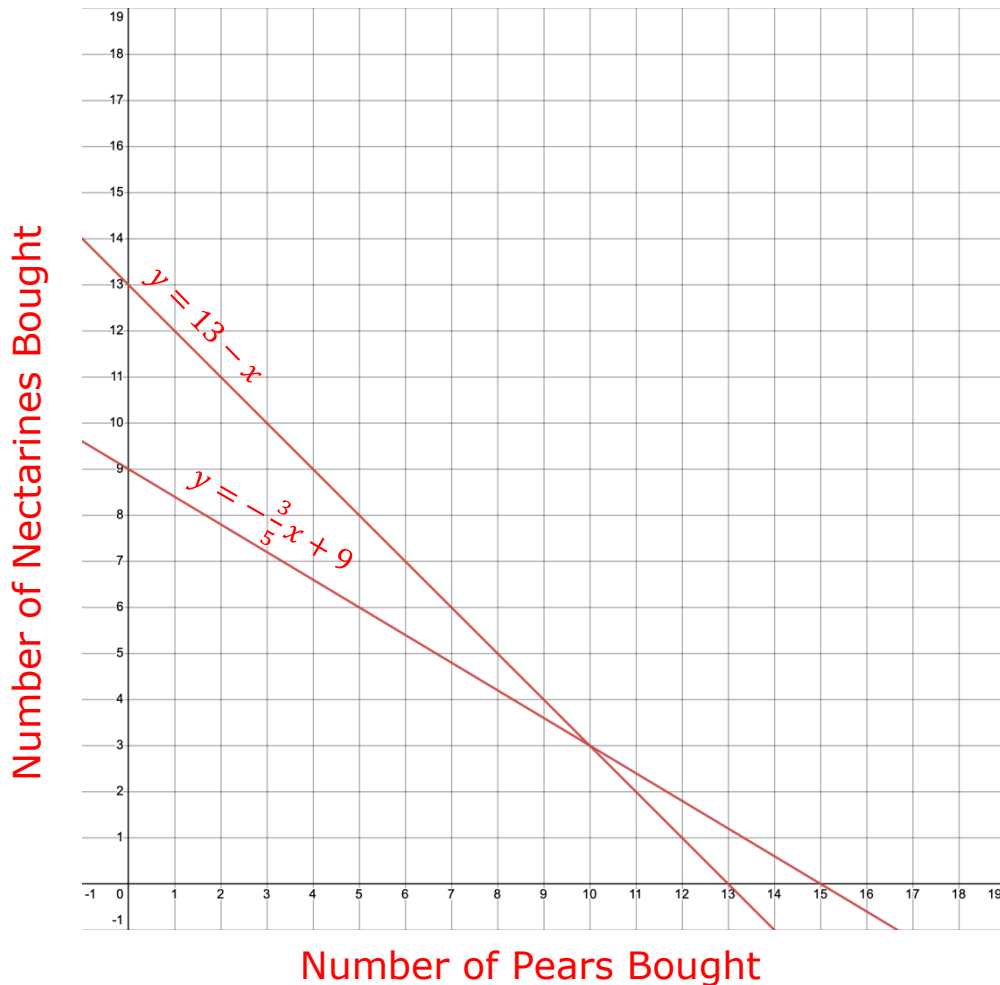


- Interpret the meaning of the solution in the context.

Jake has 5 quarters and 7 dimes.

Xander went to the grocery store and bought pears for \$0.75 each and nectarines for \$1.25 each. He bought a total of 13 pieces of fruit and spent \$11.25. The system of equations $\begin{cases} y = 13 - x \\ y = -\frac{3}{5}x + 9 \end{cases}$ can be used represents the situation where x is the number of pears and y is the number of nectarines bought.

8. Graph the system of equations. Label each axis with the fruit it represents.



9. What is the solution to the system?
(10,3)
10. Interpret the meaning of the solution in the context.
Xander bought 10 pears and 3 nectarines.